

The reply must leave before the room goes dark.

A socket failure examined three ways: as it was recorded, as it was tested, and as it was finally understood.

Entry • LN-0248

State • witnessed

Confidence • measured

Reconstruction • none

Synthesized specimen • revised presentation 2026-09-06

People, times and results illustrate the method. Original CSVs are excerpts; complete aggregate groups were constructed from the narrative on 2026-09-06. Read the source-coverage supplement.

Later corrected • 2026-09-04

LN-0249 / COR-0249-01 narrows the mechanism to write completion. The lifecycle conclusion remains supported; the instrumentation question is closed there.

3/3
IMMEDIATE STOPS FAILED

3
ATTRIBUTED OBSERVATIONS

30/30
DEFERRED STOPS PASSED

2
QUESTIONS REMAIN OPEN

SHARED PROVENANCE

00 / RECORD IDENTITY

Experiment identity

OPENED	2026-08-30T08:42:00+04:00	CLOSED	2026-08-30T11:26:00+04:00
AUTHORS	Mira Chen • AI coordinator	OPERATOR	AI coordinator
OBSERVER	Jules Okafor	WITNESS	Mira Chen
ENVIRONMENT	Linux • Node 25.2.1 • Bare 1.31.0	HANDLING	INTERNAL • SYNTHETIC IDS
CAPTURE MODE	Contemporaneous example	CONFIDENCE	MEASURED

AS RECORDED

01 / FAST FIELD NOTES

Trace notes and test choices

08:47	baseline written before tracing; expect ack then EOF
09:18	empty response again. trace looks backwards: close 1 ms before encode

09:24	do not call it causal yet—instrument timestamps may be misleading
09:51	discriminating seam: barrier after write callback, not a sleep
10:49	deferral is evidence of ordering bug, not final contract

AS RECORDED

02 / BENCH RECORD

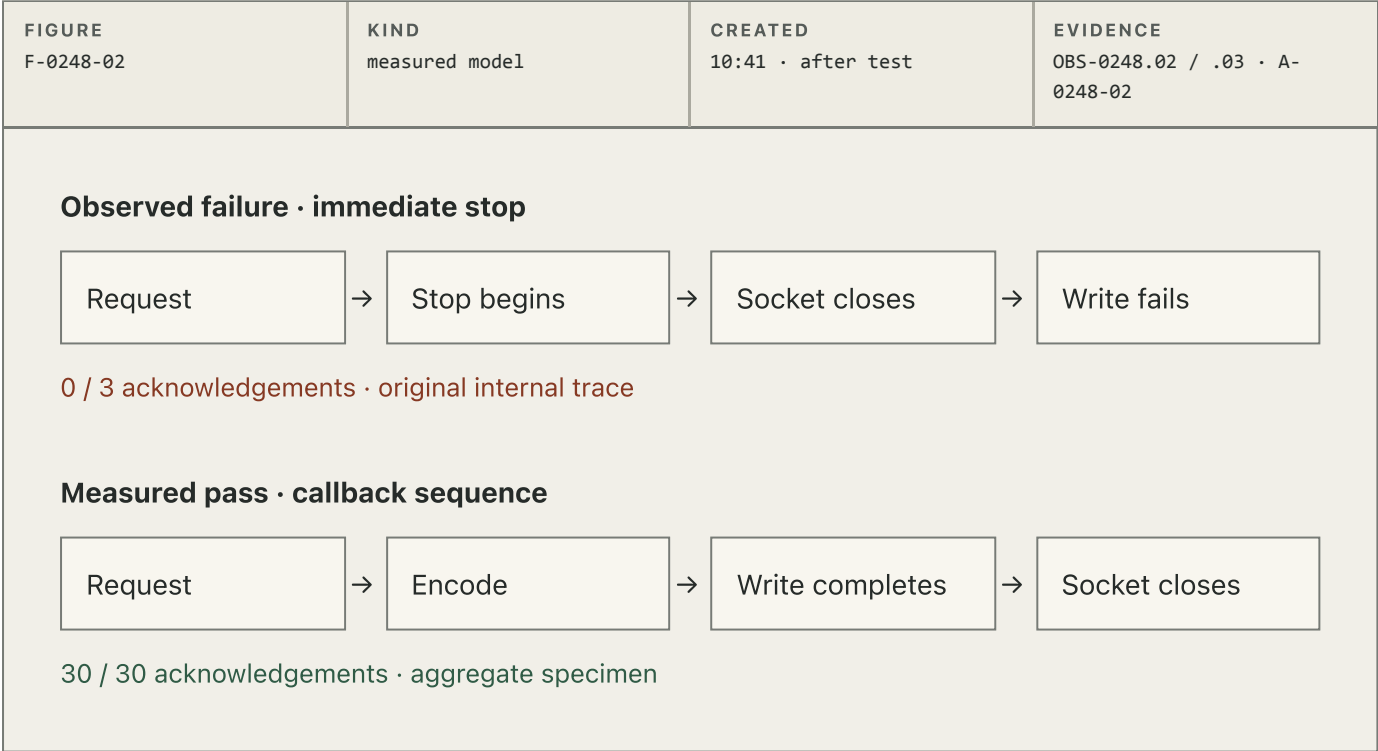
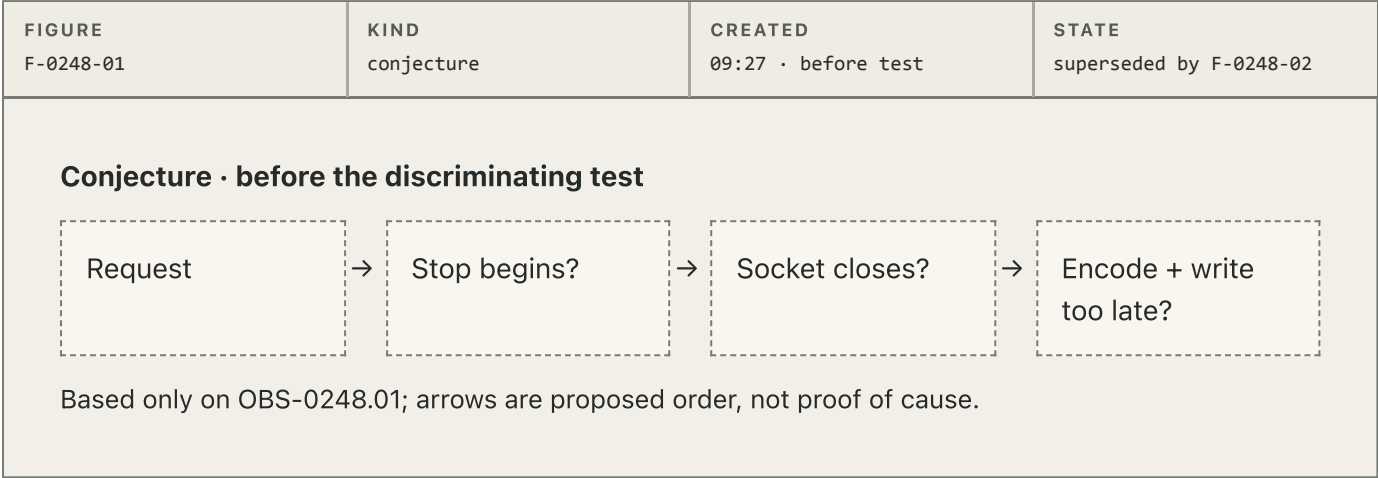
Three shutdown-order observations

OBS-0248.0109:18:14 · Mira / AI / Jules	
QUESTION What does the owner observe during current stop?	SETUP One owner socket; request 41; five trace hooks; no delay.
RAW OBSERVATION client: empty admin response trace: stop.begin → socket.close → reply.encode → failed write	
INTERPRETATION AT THE TIME Shutdown ordering is plausible; timestamp precision may mislead.	STATE / NEXT MOVE TENTATIVE Control reply completion rather than sleep.

OBS-0248.0209:51 · AI operator	
HYPOTHESIS BEFORE RUN A reply-write barrier will expose the acknowledgement if close is racing completion.	DISCRIMINATING SETUP Release test-only stop barrier from write callback; payload unchanged.
RAW OBSERVATION 10 / 10 complete acknowledgements close occurs after callback	INTERPRETATION Supports ordering race; encoding defect no longer primary.

OBS-0248.0310:36 · witnessed group	
PREDICTION BEFORE RUN One-turn deferral will pass, but does not express the protocol guarantee.	RAW MEASUREMENT immediate 0 / 3 deferred 30 / 30 callback sequence 30 / 30
INTERPRETATION Deferral demonstrates the race and repairs it within scope.	STATE / NEXT MOVE MEASURED Keep explicit lifecycle API open.

From suspected race to observed sequence



Reply completion precedes shutdown

EPISTEMIC STATE	STATEMENT
Observed	Immediate shutdown produced an empty response; close preceded successful write.
Interpreted	Closing the owner socket before reply completion was the likely cause.

EPISTEMIC STATE	STATEMENT
Discriminated	A write barrier exposed the acknowledgement without changing encoding.
Concluded	Shutdown must follow acknowledgement completion.
Projected	An explicit service lifecycle API is preferable to scheduling deferral.

CONCLUSION

The deferral is accepted as the bounded repair. It is not represented as the ideal lifecycle contract.

ATTACHED PHENOMENON

05 / ARTIFACT MANIFEST

Traces and aggregate evidence

ID	PARENT	KIND	ARTIFACT	CAPTURED BY	INTEGRITY	HANDLING
A-0248-01	OBS-0248.01	Raw trace	stop-trace.txt	AI coordinator · trace hook	Specimen / unhashed	INTERNAL
A-0248-02	OBS-0248.03	Measurement excerpt	stop-ordering.csv	Mira Chen · harness export	Specimen / unhashed	Public-safe
ART-0248-AGG	OBS-0248.01-03	Aggregate fixture	All narrative groups	Codex · constructed 2026-09-06	Synthesized / unhashed	Public specimen

DURABLE UNCERTAINTY

06 / OPEN QUESTIONS

Lifecycle and instrumentation questions

Q-0248-01 · OPEN Should reply-then-close become an explicit primitive? Needs API design and negative close/re-entry tests. Raised from OBS-0248.03.	Q-0248-02 · OPEN Can trace hooks perturb sub-millisecond ordering? Needs an external socket observer. Raised from OBS-0248.01.
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corrected by · LN-0249 · COR-0249-01

Signed, with two questions still open.

CONCLUSION BOUNDED REPAIR SUPPORTED	TIMESTAMP 2026-08-30T11:26:00+04:00
INDEPENDENT OBSERVATION Jules Okafor · R-0248-A/B	LATER RECORD LN-0249 · correction

Signed · Mira Chen + AI research coordinator

Canonical narrative · Markdown